NET (BMEVITMAC09): three mock midterms with solutions

Mock papers written by us, not official papers. Format (6 open questions, 5 points each, 30 points, pass mark 12/30, on paper) and scope (sessions 1 to 6, slides hta1 to hta6p) as announced by the lecturer and reported by Elisha. Real midterm: Thursday 2026-10-29, 8:15, IE218. Write your answers in the boxes; the solutions part repeats each question with the box filled in, followed by the marking scheme and the usual pitfalls.

Paper A

6 questions, 5 points each, 30 points, pass mark 12/30. Justify every answer.

1. IPv4 addressing (5 points)

inspired by: 2018 midterm, "same subnet"; hta2p p. 12

A host is configured as 172.20.83.140/21.

(a) Give the netmask in dotted decimal notation. (1 pt)

(b) Compute the network address, the broadcast address and the number of assignable host addresses. (2 pts)

(c) For each address below, say whether it is valid, whether it is on the same subnet as the host, and whether it can be assigned to another host of that subnet: 172.20.86.17; 172.20.79.254; 172.20.88.1; 172.20.87.255; 172.20.80.0; 172.20.84.300. (2 pts)

2. Longest prefix match (5 points)

inspired by: 2018 midterm, routing table; hta2p p. 13 and 14

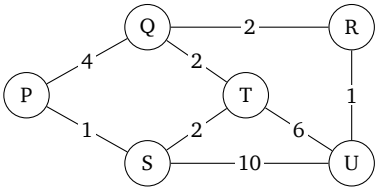
Router R uses the routing table on the left.

Prefix	Interface
84.112.0.0/12	eth0
84.120.0.0/13	eth1
84.124.0.0/14	eth2
84.126.64.0/18	eth3
0.0.0.0/0	eth4

(a) Give the outgoing interface for packets destined to 84.127.70.5, 84.126.100.1, 84.119.3.3 and 84.128.0.1. For each one, list the matching lines. (4 pts)

(b) An administrator wants to add the line 84.126.100.0/18 towards eth5. Why is it written incorrectly, and which network does it actually denote? (1 pt)

3. Link state: Dijkstra's algorithm (5 points) inspired by: 2017 topic list, no. 16; hta4p p. 8 to 14



Link costs are shown on the graph.

(a) Run Dijkstra's algorithm from P. Give a table: at each step, the node taken from the priority queue and, for every other node, the tentative distance and the previous node. On a tie, take the smallest letter. (3 pts)

(b) Derive the FIB of P: destination, cost, next hop. (1 pt)

(c) The Q-R link fails. Give the new costs from P to R and to U. According to the course, which family reconfigures faster after a link failure: link state or distance vector? (1 pt)

4. TCP Reno congestion control (5 points) inspired by: 2017 topic list, no. 33; hta5p p. 8 to 15

Conventions: cwnd and ssthresh in MSS, one round per round trip, cwnd segments sent per round. Slow Start (SS): cwnd doubles every round, **capped at ssthresh** (if doubling would exceed ssthresh, cwnd takes the value ssthresh), then Congestion Avoidance (CA). CA: cwnd +1 per round and ssthresh = cwnd. Three duplicate ACKs: cwnd and ssthresh both become half of cwnd, stay in CA. Timeout: ssthresh = cwnd/2, cwnd = 1, back to SS. A loss detected at the end of a round changes the next round; adv_wnd never limits the sender. In round 1, cwnd = 1 and ssthresh = 12. The sender receives three duplicate ACKs at the end of round 9 and has a timeout at the end of round 14.

(a) Give cwnd, ssthresh and the phase (SS or CA) for rounds 1 to 19. (3 pts)

(b) How many segments are sent in total during rounds 1 to 19? (1 pt)

(c) Why does Reno wait for three duplicate ACKs rather than one, and why does it treat a timeout more severely? (1 pt)

5. Serialization: XML to JSON and YAML (5 points) inspired by: hta1p p. 26, "Possible exam question"

A DHCP server exports a lease in the XML format shown on the left.

```
<lease>
  <ip>192.168.10.57</ip>
  <hostname>labo-pc7</hostname>
  <duration>86400</duration>
  <options>
    <router>192.168.10.1</router>
    <dns>192.168.10.2</dns>
    <dns>192.168.10.3</dns>
  </options>
</lease>
```

(a) Write the same content in JSON. (2 pts)

(b) Write it in YAML. (2 pts)

(c) Why does the course recommend `JSON.parse()` rather than `eval()` to read JSON in JavaScript? (1 pt)

6. Multiple access, duplexing and token bucket (5 points) inspired by: 2018 midterm, Q1; hta1 p. 19 to 21; hta1p p. 18 and 19

A radio operator has two 1.6 MHz bands, one for the uplink and one for the downlink. Each band is split into 200 kHz carriers, and each carrier into frames of 8 time slots. A call uses one slot on an uplink carrier and one slot on a downlink carrier.

(a) Define FDMA, TDMA and CDMA in one sentence each. Which multiple access and duplexing techniques does this operator combine? What is the maximum number of simultaneous calls? (2 pts)

(b) A second operator has a single 3.2 MHz band, with the same carriers and frames, and uses Time Division Duplex (TDD). What is the principle of TDD? What is the maximum number of symmetric calls? Which constraint does time sharing impose on the equipment? (1 pt)

(c) At the network entrance, a client's flow is metered by a token bucket (one token = one bit). Contract: average bitrate 4 Mbit/s, maximum bitrate 10 Mbit/s, bursts of 0.3 s. Give K and B with the design rule of the course. The bucket is full and the client sends at 10 Mbit/s: after how long is the bucket empty, and what happens then to the traffic above 4 Mbit/s? What would a leaky bucket do instead? (2 pts)

1. Variable length subnetting (5 points) inspired by: 2019 practice set, ex. 6 (splitting a /20); hta2p p. 7

A company owns the block 10.62.96.0/20. Needs: Production 1000 hosts, Offices 450, Laboratory 120, Guest Wi-Fi 60, and two point-to-point links between routers (two addresses each).

(a) Which prefix length does each need require? (1 pt)

(b) Allocate the subnets from the start of the block, largest first. For each one, give network/prefix and broadcast address. (3 pts)

(c) What is the largest aligned subnet still free in the block? Which single prefix does the company announce to its provider? (1 pt)

2. CIDR aggregation (5 points) inspired by: 2018 midterm, Q10; hta2p p. 14

A router has the table on the left and no default route.

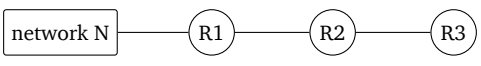
Prefix	Next hop
185.40.16.0/22	A
185.40.20.0/22	A
185.40.24.0/21	A
185.40.32.0/21	A
185.40.40.0/22	B
185.40.44.0/22	A
185.40.48.0/20	A

(a) Give the address range of each line (the 3rd octet is enough). (1 pt)

(b) Reduce the table to the minimum number of lines by exact merging: the new table covers exactly the same addresses, every address keeps its next hop, and no line may overlap another. (2.5 pts)

(c) Now a more specific line may override a wider one (longest prefix match). Reduce the table to three lines without changing the output of any address. Why can't everything be summarized as 185.40.0.0/18 towards A? (1.5 pts)

3. Distance vector: count to infinity (5 points) inspired by: 2018 midterm, Q3; 2020 exams; hta4p p. 4 to 6



The routers run a RIP-like distance vector protocol: hop count metric, R1 is directly connected to N with distance 1, infinity is 16.

(a) After convergence, give the distance and next hop towards N in R1, R2 and R3. What does R2's advertisement contain about N, and what is not in it? (1.5 pts)

(b) The R1-R2 link goes down. R2 marks N unreachable, but receives R3's periodic update before it has advertised the failure; updates then alternate, R3 then R2. Describe the first four updates (router, distance, next hop). After how many updates do both R2 and R3 consider N unreachable? What happens meanwhile to packets towards N? (2 pts)

(c) Explain split horizon and poisoned reverse on this example: what does R3 advertise to R2 about N in each case, and what becomes of scenario (b)? (1.5 pts)

4. NAT (5 points) inspired by: no past paper; hta5 p. 18 and 19

Home network 192.168.0.0/24; NAT gateway with internal address 192.168.0.1 and external address 81.183.20.9. Hosts 192.168.0.23 and 192.168.0.42 each open a TCP connection to server 145.236.10.5, port 443, both with source port 51000. The gateway picks external ports 40100 (for .23) and then 40101 (for .42).

(a) For the connection of host .23, give source and destination (address and port) on the four legs: host to gateway, gateway to server, server to gateway, gateway to host. (2 pts)

(b) Write the gateway's association table. Why can't the gateway keep port 51000 for both connections? Reason with the 5-tuple. (1.5 pts)

(c) An internal web server, 192.168.0.50 port 8080, must be reachable from the Internet. Why doesn't the NAT described here allow it, and which solutions does the course mention? (1.5 pts)

5. Serialization: XML to JSON and YAML (5 points) inspired by: hta1p p. 26, "Possible exam question"

A router configuration is given in XML on the left.

```
<router>
  <name>edge1</name>
  <ospfArea>0.0.0.0</ospfArea>
  <interfaces>
    <interface><name>eth0</name>
      <address>10.0.0.1/30</address>
      <mtu>1500</mtu></interface>
    <interface><name>eth1</name>
      <address>192.168.5.1/24</address>
      <mtu>9000</mtu></interface>
  </interfaces>
  <protocols>
    <protocol>ospf</protocol>
    <protocol>bgp</protocol>
  </protocols>
</router>
```

(a) Write the same content in JSON. (2 pts)

(b) Write it in YAML. (2 pts)

(c) The course also presents TLV (Type, Length, Value) encoding. Why can a reader written for an older version of a TLV protocol still read a message that contains a new type? (1 pt)

6. Web: HTTP, TLS and QUIC; quality of service (5 points) inspired by: no past paper; hta6 p. 8 to 15; hta1p p. 16

A browser opens `https://www.example.org/index.html` for the first time.

(a) List in order the exchanges needed until the page is received with HTTP/1.1: protocol, transport, port, role of each exchange. (2 pts)

(b) The server presents an X.509 certificate. Under which two conditions does the browser accept it? What does a self-signed certificate allow, and what does it not allow? (1 pt)

(c) What changes with HTTP/3 over QUIC: transport, handshakes, encryption, mobility? (1 pt)

(d) Outside the Web, session 1: the course distinguishes IntServ, DiffServ and best effort. How do they differ? Which one describes how the Internet runs, and why? (1 pt)

1. Session 1: Protocol stack and statistical multiplexing (5 points) inspired by: hta1p p. 3 to 6 and 20; hta5 p. 7

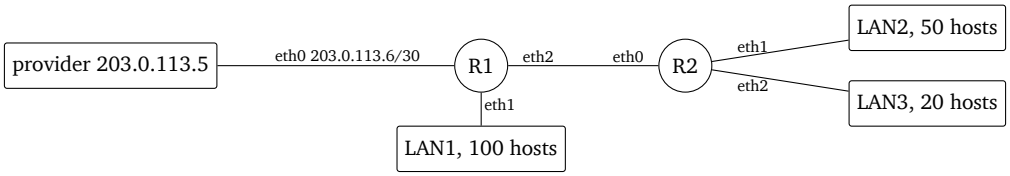
A browser sends an HTTP request; it goes down through TCP, IPv4 and Ethernet.

(a) Name the 7 OSI layers and the 4 TCP/IP layers and match them. Give the PDU name at the transport, network, datalink and physical layers. (1.5 pts)

(b) In which order do the headers appear in the transmitted frame? What is encapsulation, and why is the UDP checksum, which covers fields of the IP header, called a layering violation? (1 pt)

(c) Statistical multiplexing: $N = 400$ subscribers share a multiplexer; each is active with probability $p = 2\%$ at a given time, independently. What is the average number of active subscribers? Which distribution does the number X of active subscribers follow, and which approximation does the course use? With Chebyshev's inequality for $k = 3$ (take $\sigma = \sqrt{\lambda}$ for the Poisson distribution), which capacity L (an integer number of links) guarantees a blocking probability $\Pr(X > L)$ of at most 1/9? The exact Poisson computation gives $\Pr(X > L) \approx 0.16\%$ for that L: what do you conclude about Chebyshev? (2.5 pts)

2. Session 2: Interface addresses and routing tables (5 points) inspired by: 2018 midterm, Q7; 2019 practice set, ex. 1



The company owns the block 192.168.40.0/24 for LAN1, LAN2, LAN3 and the R1-R2 link.

(a) Split the block from its start, largest need first: network/prefix of LAN1, LAN2, LAN3 and of the link. (1.5 pts)

(b) Give an address to every interface of R1 and R2 on these networks, and the default gateway of a PC on LAN3. (1 pt)

(c) Write the routing tables of R1 and R2 (destination/prefix, next hop or interface), including the default route. (1.5 pts)

(d) Can R1 replace its routes to LAN2 and LAN3 with the single line 192.168.40.128/25? Follow a packet destined to 192.168.40.240. Why must R1 do NAT on eth0? (1 pt)

3. Session 3: DHCP, DNS and ICMP (5 points) inspired by: no past paper; hta3 p. 4 to 17; hta3p p. 18 to 20

A laptop with no network configuration joins a local network; the user then runs `traceroute www.example.org`.

(a) Describe the DHCP exchange: the four messages in order, which ones are broadcast and why, the transport and the ports, three pieces of information supplied besides the address. What happens if the lease is not renewed? (2 pts)

(b) How does the laptop obtain the IPv4 address of www.example.org: which server does it ask and how does it know that server, which record does it request, where can the answer come from? (1.5 pts)

(c) Explain how traceroute discovers the routers on the path: IP header field used, ICMP messages involved. (1.5 pts)

4. Session 4: One Bellman-Ford iteration (distance vector) (5 points) inspired by: 2017 topic list, no. 15; hta4p p. 3 and 4

Vector from	A	B	C	D	E	F
A (cost 2)	0	4	6	5	8	10
B (cost 7)	4	0	2	1	3	9
C (cost 3)	6	2	0	7	2	4

Router X has three neighbours, A, B and C, with the link costs shown. It has just received their distance vectors (left, destinations other than X).

(a) Compute X's new table from these vectors only: for each destination A to F, distance and next hop, with the formula used. (2.5 pts)

(b) Which vector does X advertise next? With split horizon, what does X's advertisement to C contain? (1 pt)

(c) Compare distance vector and link state (information sent, recipients, algorithm, convergence), with one protocol of each family. Which protocol is used between autonomous systems, and of which type is it? (1.5 pts)

5. Session 5: TCP connection, fast retransmit and window size (5 points) inspired by: 2020 exams; hta5 p. 9, 10; hta5p p. 13, 16

(a) A client opens a TCP connection with initial sequence number $S_c = 7300$; the server picks $S_s = 1200$. Give the flags, Sequence and Acknowledgement numbers of the three handshake segments. Which attack targets the server while it waits for the last segment, and which defense does the course give? (1.5 pts)

(b) As on the slides, numbers count segments. The sender sends segments 40 to 45; segment 42 is lost; the receiver acknowledges every segment it receives. List the ACKs. When does Reno retransmit, which ACK follows the arrival of the retransmitted 42, and what does Reno do to cwnd? (1.5 pts)

(c) A 200 Mbit/s link has a 60 ms RTT. Which window, in octets, is needed to use it fully? Which maximum throughput does the Window Size field alone allow (16 bits, counted in octets)? Which minimum window scale shift is needed? (2 pts)

6. Session 6: E-mail (5 points) inspired by: no past paper; hta6p p. 4 to 18

alice@example.org sends, from her mail client, a message with a PDF attachment to bob@example.net, who reads his mail on his phone and on his computer.

(a) Describe the path of the message: agents crossed (MUA, MTA, MDA), protocol and port at each step, and how the MTA of example.org finds the one of example.net. (2 pts)

(b) Why should Bob use IMAP rather than POP3? (1 pt)

(c) The message format only allows 7-bit ASCII: how is the attachment carried? (1 pt)

(d) How can the MTA of example.net check that the message comes from a server authorized for example.org? Name two mechanisms and their principle. (1 pt)

Solutions

Solutions and indicative marking schemes written by us, 5 points per question. According to the lecturer's remarks reported by Elisha, marking is lenient: an approximate explanation that goes in the right direction earns points. All calculations are checked by `calc_examens_blancs_net.py`.

Paper A: solutions

6 questions, 5 points each, 30 points, pass mark 12/30. Justify every answer.

1. IPv4 addressing (5 points)

inspired by: 2018 midterm, "same subnet"; hta2p p. 12

A host is configured as 172.20.83.140/21.

(a) Give the netmask in dotted decimal notation. (1 pt)

/21 = 8 + 8 + 5: 255.255.**248**.0 (248 = `11111000`).

(b) Compute the network address, the broadcast address and the number of assignable host addresses. (2 pts)

The interesting octet is the 3rd; block size $m = 256 - 248 = 8$. 83 = `01010011`; the largest multiple of 8 not above 83 is 80. Network **172.20.80.0/21**, broadcast **172.20.87.255**, $2^{11} - 2 =$ **2046** hosts (172.20.80.1 to 172.20.87.254).

(c) For each address below, say whether it is valid, whether it is on the same subnet as the host, and whether it can be assigned to another host of that subnet: 172.20.86.17; 172.20.79.254; 172.20.88.1; 172.20.87.255; 172.20.80.0; 172.20.84.300. (2 pts)

Address	Valid	Same subnet	Assignable
172.20.86.17	yes	yes (86 is in 80 to 87)	yes
172.20.79.254	yes	no (network 172.20.72.0/21)	no
172.20.88.1	yes	no (network 172.20.88.0/21)	no
172.20.87.255	yes	yes	no: broadcast address
172.20.80.0	yes	yes	no: network address
172.20.84.300	no: 300 > 255		

Marking. (a) 1. (b) network 1, broadcast and host count 1. (c) 2, minus 0.5 per wrong row. **Pitfalls.** Forgetting the −2; assuming 172.20.79.254 is a neighbour because it is numerically close; calling the broadcast and network addresses assignable because they are on the subnet.

2. Longest prefix match (5 points)

inspired by: 2018 midterm, routing table; hta2p p. 13 and 14

Router R uses the routing table on the left.

Prefix	Interface
84.112.0.0/12	eth0
84.120.0.0/13	eth1
84.124.0.0/14	eth2
84.126.64.0/18	eth3
0.0.0.0/0	eth4

(a) Give the outgoing interface for packets destined to 84.127.70.5, 84.126.100.1, 84.119.3.3 and 84.128.0.1. For each one, list the matching lines. (4 pts)

Ranges: /12: 84.112.0.0 to 84.127.255.255; /13: 84.120 to 84.127; /14: 84.124 to 84.127; /18: 84.126.64.0 to 84.126.127.255.

Destination	Matching lines	Interface
84.127.70.5	/12, /13, /14, /0 (not the /18: 2nd octet is 127, not 126)	**eth2**
84.126.100.1	/12, /13, /14, /18 (100 is in 64 to 127), /0	**eth3**
84.119.3.3	/12, /0 (119 < 120, outside the /13)	**eth0**
84.128.0.1	/0 only	**eth4**

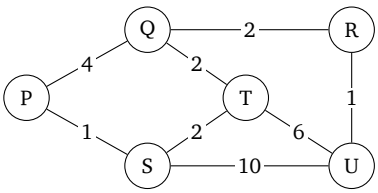
(b) An administrator wants to add the line 84.126.100.0/18 towards eth5. Why is it written incorrectly, and which network does it actually denote? (1 pt)

In a /18 the 3rd octet of the network is a multiple of 64, and 100 is not. The line actually denotes 84.126.64.0/18 (100 = `01100100`, keep `01`), which already exists towards eth3: conflict.

Marking. (a) 1 per destination, 0.5 of it for the justification. (b) 1. **Pitfalls.** 84.127.70.5 has its 3rd octet in 64 to 127 but a different 2nd octet; the default route is used only when nothing else matches.

3. Link state: Dijkstra's algorithm (5 points)

inspired by: 2017 topic list, no. 16; hta4p p. 8 to 14



Link costs are shown on the graph.

(a) Run Dijkstra's algorithm from P. Give a table: at each step, the node taken from the priority queue and, for every other node, the tentative distance and the previous node. On a tie, take the smallest letter. (3 pts)

Cell "d,p": tentative distance, previous node; bold: the node taken at that step (final).

Step	Taken	Q	R	S	T	U
0	P (0)	4,P	∞	1,P	∞	∞
1	S (1)	4,P	∞	**1**	3,S	11,S
2	T (3)	4,P	∞		**3**	9,T
3	Q (4)	**4**	6,Q			9,T
4	R (6)		**6**			7,R
5	U (7)					**7**

At step 2 the path P-S-T-Q costs 5, not better than 4: Q keeps P as previous node.

(b) Derive the FIB of P: destination, cost, next hop. (1 pt)

Q 4 via Q; R 6 via Q (P-Q-R); S 1 via S; T 3 via S (P-S-T); U 7 via Q (P-Q-R-U).

(c) The Q-R link fails. Give the new costs from P to R and to U. According to the course, which family reconfigures faster after a link failure: link state or distance vector? (1 pt)

Without Q-R: U = 9 via P-S-T-U, R = 10 via P-S-T-U-R, both via S. The slides state that LS can detect and reconfigure faster than DV after a link failure.

Marking. (a) 3 (2 if one isolated error is propagated consistently). (b) 1. (c) 0.5 costs, 0.5 comparison. **Pitfalls.** Taking the direct S-U (10) or T-U (6) link for U; updating a node that has already been taken; writing the whole path in the FIB instead of the next hop.

4. TCP Reno congestion control (5 points)

inspired by: 2017 topic list, no. 33; hta5p p. 8 to 15

Conventions: cwnd and ssthresh in MSS, one round per round trip, cwnd segments sent per round. Slow Start (SS): cwnd doubles every round, **capped at ssthresh** (if doubling would exceed ssthresh, cwnd takes the value ssthresh), then Congestion Avoidance (CA). CA: cwnd +1 per round and ssthresh = cwnd. Three duplicate ACKs: cwnd and ssthresh both become half of cwnd, stay in CA. Timeout: ssthresh = cwnd/2, cwnd = 1, back to SS. A loss detected at the end of a round changes the next round; adv_wnd never limits the sender. In round 1, cwnd = 1 and ssthresh = 12. The sender receives three duplicate ACKs at the end of round 9 and has a timeout at the end of round 14.

(a) Give cwnd, ssthresh and the phase (SS or CA) for rounds 1 to 19. (3 pts)

Round	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
cwnd	1	2	4	8	12	13	14	15	16	8	9	10	11	12	1	2	4	6	7
ssthresh	12	12	12	12	12	13	14	15	16	8	9	10	11	12	6	6	6	6	7
Phase	SS	SS	SS	SS	CA	CA	CA	CA	CA	CA	CA	CA	CA	CA	SS	SS	SS	CA	CA

Round 5: doubling would give 16, capped at 12. End of round 9: 3 duplicate ACKs with cwnd = 16, so cwnd = ssthresh = 8. End of round 14: timeout with cwnd = 12, so ssthresh = 6, cwnd = 1; round 18: 8 capped at 6. Rounds 5 and 18, where cwnd reaches ssthresh, may be marked SS or CA: what matters is that the next round grows by +1.

(b) How many segments are sent in total during rounds 1 to 19? (1 pt)

1 + 2 + 4 + 8 + 12 + 13 + 14 + 15 + 16 = 85 in rounds 1 to 9; **155** segments in total.

(c) Why does Reno wait for three duplicate ACKs rather than one, and why does it treat a timeout more severely? (1 pt)

Out-of-order delivery without loss also produces duplicate ACKs; three of them make a loss very likely. Duplicate ACKs mean that segments still get through (mild congestion: halve cwnd). A timeout means nothing gets through: the congestion looks severe, back to slow start.

Marking. (a) 3: 1 for slow start with the cap, 1 for fast recovery, 1 for the timeout. (b) 1. (c) 1. **Pitfalls.** Doubling to 16 in round 5 without applying the convention; going back to 1 after duplicate ACKs (that is Tahoe); halving the initial ssthresh (12) instead of cwnd at the time of the loss.

5. Serialization: XML to JSON and YAML (5 points)

inspired by: hta1p p. 26, "Possible exam question"

A DHCP server exports a lease in the XML format shown on the left.

```
<lease>
  <ip>192.168.10.57</ip>
  <hostname>labo-pc7</hostname>
  <duration>86400</duration>
  <options>
    <router>192.168.10.1</router>
    <dns>192.168.10.2</dns>
    <dns>192.168.10.3</dns>
  </options>
</lease>
```

(a) Write the same content in JSON. (2 pts)

Slide convention (hta1p p. 24 to 26): the root element disappears and its content becomes the top-level object; repeated siblings become an array. Keeping the root as a key (`"lease": {…}`) is also consistent; we accept it.

```
{
  "ip": "192.168.10.57",
  "hostname": "labo-pc7",
  "duration": 86400,
  "options": {
    "router": "192.168.10.1",
    "dns": ["192.168.10.2", "192.168.10.3"]
  }
}
```

(b) Write it in YAML. (2 pts)

```
ip: 192.168.10.57
hostname: labo-pc7
duration: 86400
options:
  router: 192.168.10.1
  dns:
    - 192.168.10.2
    - 192.168.10.3
```

(c) Why does the course recommend `JSON.parse()` rather than `eval()` to read JSON in JavaScript? (1 pt)

`eval()` runs the text as JavaScript code: a malicious text would execute whatever it contains. `JSON.parse()` only reads data, so it is more secure.

Marking. (a) 2: structure 1, `dns` array 0.5, number without quotes and addresses quoted 0.5. (b) 2: indentation 1, hyphenated list 1. (c) 1. **Pitfalls.** Two `"dns"` keys in the same object (the second overwrites the first); quoting 86400; losing the indentation under `options`.

6. Multiple access, duplexing and token bucket (5 points)

inspired by: 2018 midterm, Q1; hta1 p. 19 to 21; hta1p p. 18 and 19

A radio operator has two 1.6 MHz bands, one for the uplink and one for the downlink. Each band is split into 200 kHz carriers, and each carrier into frames of 8 time slots. A call uses one slot on an uplink carrier and one slot on a downlink carrier.

(a) Define FDMA, TDMA and CDMA in one sentence each. Which multiple access and duplexing techniques does this operator combine? What is the maximum number of simultaneous calls? (2 pts)

FDMA: each source mixes with a different carrier. TDMA: each source has a time slot for sending, which needs carrier sense or time synchronization. CDMA: each source uses a different spreading code, on the same band at the same time; the codes cannot be fully orthogonal if the sources are not perfectly synchronized. The operator combines FDMA and TDMA (carriers split into slots) with Frequency Division Duplex (FDD: one band per direction). 1.6 MHz/200 kHz = 8 carriers per direction, ×8 slots: **64** calls.

(b) A second operator has a single 3.2 MHz band, with the same carriers and frames, and uses Time Division Duplex (TDD). What is the principle of TDD? What is the maximum number of symmetric calls? Which constraint does time sharing impose on the equipment? (1 pt)

TDD: both directions use the same frequency and take turns. 3.2 MHz/200 kHz = 16 carriers ×8 = 128 slots; a call needs two (one per direction): **64** calls, as many as with FDD and the same total bandwidth. Constraint: the equipment must be time synchronized to know when to send and when to receive.

(c) At the network entrance, a client's flow is metered by a token bucket (one token = one bit). Contract: average bitrate 4 Mbit/s, maximum bitrate 10 Mbit/s, bursts of 0.3 s. Give K and B with the design rule of the course. The bucket is full and the client sends at 10 Mbit/s: after how long is the bucket empty, and what happens then to the traffic above 4 Mbit/s? What would a leaky bucket do instead? (2 pts)

K = average bitrate = 4 Mbit/s ($4 \cdot 10^6$ tokens/s). $B = (\text{max} - \text{avg}) \times \text{burstlen} = (10 - 4) \times 0.3 =$ **1.8 Mbit**. The bucket loses $10 - 4 = 6$ Mbit/s net, so it is empty after $1.8/6 =$ **0.3 s**, the contracted burst length. Then only the messages covered by new tokens pass (4 Mbit/s); the excess is dropped. A leaky bucket shapes the flow: output at most K, the excess waits in a queue and is dropped only when the bucket is full.

Marking. (a) 2: definitions 1, FDMA, TDMA and FDD named 0.5, 64 calls 0.5. (b) 1: principle and synchronization 0.5, 64 calls 0.5. (c) 2: K and B 1, 0.3 s and fate of the excess 0.5, leaky bucket 0.5. **Pitfalls.** Answering that FDMA and TDD cannot work together (they combine); stopping at the definitions without describing the combination; counting 128 calls in TDD and forgetting one slot per direction; mixing up metering (token bucket, drop) and shaping (leaky bucket, queue).

1. Variable length subnetting (5 points) inspired by: 2019 practice set, ex. 6 (splitting a /20); hta2p p. 7

A company owns the block 10.62.96.0/20. Needs: Production 1000 hosts, Offices 450, Laboratory 120, Guest Wi-Fi 60, and two point-to-point links between routers (two addresses each).

(a) Which prefix length does each need require? (1 pt)

Smallest h with $2^h - 2 \geq$ need: 1000 gives /22 (1022); 450 gives /23 (510); 120 gives /25 (126); 60 gives /26 (62); links /30 (2).

(b) Allocate the subnets from the start of the block, largest first. For each one, give network/prefix and broadcast address. (3 pts)

Need	Network/prefix	Broadcast	Hosts
Production	10.62.96.0/22	10.62.99.255	1022
Offices	10.62.100.0/23	10.62.101.255	510
Laboratory	10.62.102.0/25	10.62.102.127	126
Guest Wi-Fi	10.62.102.128/26	10.62.102.191	62
Link 1	10.62.102.192/30	10.62.102.195	2
Link 2	10.62.102.196/30	10.62.102.199	2

Each block starts on a multiple of its size (100 is even for the /23, 102.128 is a multiple of 64): allocating in decreasing order guarantees alignment.

(c) What is the largest aligned subnet still free in the block? Which single prefix does the company announce to its provider? (1 pt)

Free: 10.62.102.200 to 10.62.111.255, 2360 addresses. Largest aligned block: **10.62.104.0/21** (104 to 111). Single announcement: **10.62.96.0/20**.

Marking. (a) 1. (b) 3, 0.5 per row. (c) 0.5 and 0.5. **Pitfalls.** /23 for 1000 hosts ($2^9 - 2 = 510$); /27 for the guests ($2^5 - 2 = 30$); starting with the small blocks, which breaks the alignment of the large ones.

2. CIDR aggregation (5 points) inspired by: 2018 midterm, Q10; hta2p p. 14

A router has the table on the left and no default route.

Prefix	Next hop
185.40.16.0/22	A
185.40.20.0/22	A
185.40.24.0/21	A
185.40.32.0/21	A
185.40.40.0/22	B
185.40.44.0/22	A
185.40.48.0/20	A

(a) Give the address range of each line (the 3rd octet is enough). (1 pt)

3rd octet: 16 to 19; 20 to 23; 24 to 31; 32 to 39; 40 to 43 (B); 44 to 47; 48 to 63.

(b) Reduce the table to the minimum number of lines by exact merging: the new table covers exactly the same addresses, every address keeps its next hop, and no line may overlap another. (2.5 pts)

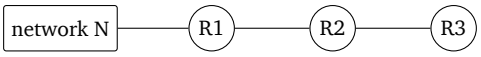
16/22 + 20/22: contiguous, same size, 16 is a multiple of 8, so 16.0/21; + 24/21: 16 is a multiple of 16, so **185.40.16.0/20**. 32/21 could only merge with 40/21, half of which (40 to 43) goes to B. 44/22 could only merge with 40/22 (B). 16/20 and 32/21 differ in size, and 16/20 would only merge with 0/20 (absent). Result, 5 lines: 185.40.16.0/20 A; 185.40.32.0/21 A; 185.40.40.0/22 B; 185.40.44.0/22 A; 185.40.48.0/20 A.

(c) Now a more specific line may override a wider one (longest prefix match). Reduce the table to three lines without changing the output of any address. Why can't everything be summarized as 185.40.0.0/18 towards A? (1.5 pts)

185.40.16.0/20 A; **185.40.32.0/19** A (32 to 63); 185.40.40.0/22 B. The /22 towards B is more specific and wins over the /19 for 40 to 43; every other address keeps A. 185.40.0.0/18 would also cover 185.40.0.0 to 185.40.15.255, which are not in the table: these packets, dropped today for lack of a route, would go to A.

Marking. (a) 1. (b) 2.5: 16.0/20 1.5, other merges correctly refused 1. (c) 1.5: three lines 1, /18 refused 0.5. **Pitfalls.** Merging 32/21 and 40/22 although 40/22 goes to B; writing 185.40.16.0/19 (16 is not a multiple of 32); aggregating wider than the addresses present.

3. Distance vector: count to infinity (5 points) inspired by: 2018 midterm, Q3; 2020 exams; hta4p p. 4 to 6



The routers run a RIP-like distance vector protocol: hop count metric, R1 is directly connected to N with distance 1, infinity is 16.

(a) After convergence, give the distance and next hop towards N in R1, R2 and R3. What does R2's advertisement contain about N, and what is not in it? (1.5 pts)

R1: 1, directly connected; R2: 2 via R1; R3: 3 via R2. R2 advertises (N, 2): destination and distance, without the next hop. So R3 does not know that R2's route goes through R1.

(b) The R1-R2 link goes down. R2 marks N unreachable, but receives R3's periodic update before it has advertised the failure; updates then alternate, R3 then R2. Describe the first four updates (router, distance, next hop). After how many updates do both R2 and R3 consider N unreachable? What happens meanwhile to packets towards N? (2 pts)

1. R2 receives "N at 3" from R3 and sets N to 4 via R3 (it does not know this route goes through itself). 2. R3 receives "N at 4" from its next hop R2 and moves to 5 via R2: a router always follows its next hop, even if the distance grows. 3. R2 moves to 6 via R3. 4. R3 moves to 7 via R2. After k updates the distance is $k + 3$: R2 reaches 16 at the 13th, R3 at the **14th**. Meanwhile, packets towards N bounce between R2 and R3 until their TTL expires.

(c) Explain split horizon and poisoned reverse on this example: what does R3 advertise to R2 about N in each case, and what becomes of scenario (b)? (1.5 pts)

Split horizon: R3 does not advertise N to R2, its next hop for N; R2 keeps N unreachable and advertises that to R3, which marks it unreachable in turn. Poisoned reverse: R3 advertises N with an infinite cost (16) to R2, which will not choose R3 as next hop. Either way, no counting. Without a fix, RIP's hop count limit of 15 at least bounds the counting.

Marking. (a) 1.5. (b) 2: mechanism 1, numbers 0.5, looping packets 0.5. (c) 1.5. **Pitfalls.** Forgetting that R3 accepts a larger distance because it comes from its next hop; saying that the advertisement carries the path (that is BGP, path vector).

4. NAT (5 points)

inspired by: no past paper; hta5 p. 18 and 19

Home network 192.168.0.0/24; NAT gateway with internal address 192.168.0.1 and external address 81.183.20.9. Hosts 192.168.0.23 and 192.168.0.42 each open a TCP connection to server 145.236.10.5, port 443, both with source port 51000. The gateway picks external ports 40100 (for .23) and then 40101 (for .42).

(a) For the connection of host .23, give source and destination (address and port) on the four legs: host to gateway, gateway to server, server to gateway, gateway to host. (2 pts)

Leg	Source	Destination
host to gateway	192.168.0.23:51000	145.236.10.5:443
gateway to server	81.183.20.9:40100	145.236.10.5:443
server to gateway	145.236.10.5:443	81.183.20.9:40100
gateway to host	145.236.10.5:443	192.168.0.23:51000

(b) Write the gateway's association table. Why can't the gateway keep port 51000 for both connections? Reason with the 5-tuple. (1.5 pts)

External port 40100 ↔ 192.168.0.23:51000; 40101 ↔ 192.168.0.42:51000 (TCP, towards 145.236.10.5:443). With 51000 for both, the two connections would have the same 5-tuple on the Internet side, {81.183.20.9, 145.236.10.5, TCP, 51000, 443}: the gateway could not tell which host a reply belongs to. A distinct external port per connection makes each 5-tuple unique.

(c) An internal web server, 192.168.0.50 port 8080, must be reachable from the Internet. Why doesn't the NAT described here allow it, and which solutions does the course mention? (1.5 pts)

Source NAT (S-NAT) only works for connections initiated from the private network: a SYN arriving at 81.183.20.9:8080 matches no association and cannot be forwarded. Solutions: D-NAT, a fixed rule mapping an external port to 192.168.0.50:8080; or NAT traversal, where the application asks the gateway to open a port for incoming connections (UPnP Internet Gateway Protocol).

Marking. (a) 2, 0.5 per leg. (b) 1.5: table 0.5, 5-tuple 1. (c) 1.5. **Pitfalls.** Rewriting the destination instead of the source; forgetting the reverse rewriting on the way back.

5. Serialization: XML to JSON and YAML (5 points)

inspired by: hta1p p. 26, "Possible exam question"

A router configuration is given in XML on the left.

```
<router>
  <name>edge1</name>
  <ospfArea>0.0.0.0</ospfArea>
  <interfaces>
    <interface><name>eth0</name>
      <address>10.0.0.1/30</address>
      <mtu>1500</mtu></interface>
    <interface><name>eth1</name>
      <address>192.168.5.1/24</address>
      <mtu>9000</mtu></interface>
  </interfaces>
  <protocols>
    <protocol>ospf</protocol>
    <protocol>bgp</protocol>
  </protocols>
</router>
```

(a) Write the same content in JSON. (2 pts)

```
{
  "name": "edge1",
  "ospfArea": "0.0.0.0",
  "interfaces": [
    { "name": "eth0", "address": "10.0.0.1/30", "mtu": 1500 },
    { "name": "eth1", "address": "192.168.5.1/24", "mtu": 9000 }
  ],
  "protocols": ["ospf", "bgp"]
}
```

The repeated `<interface>` elements become an array of objects, the `<protocol>` elements an array of strings; the repeated tags disappear.

(b) Write it in YAML. (2 pts)

```
name: edge1
ospfArea: 0.0.0.0
interfaces:
  - name: eth0
    address: 10.0.0.1/30
    mtu: 1500
```

```
  - name: eth1
    address: 192.168.5.1/24
    mtu: 9000
protocols:
  - ospf
  - bgp
```

(c) The course also presents TLV (Type, Length, Value) encoding. Why can a reader written for an older version of a TLV protocol still read a message that contains a new type? (1 pt)

Every element carries its length: the reader skips the value of an unknown type and goes on.

Marking. (a) 2: array of objects 1, rest 1. (b) 2: hyphens and alignment of the keys of one element 1, rest 1. (c) 1. **Pitfalls.** Writing `0.0.0.0` without quotes in JSON (it is not a number); in YAML, a hyphen before every key of an interface instead of one hyphen per list element.

6. Web: HTTP, TLS and QUIC; quality of service (5 points)

inspired by: no past paper; hta6 p. 8 to 15; hta1p p. 16

A browser opens `https://www.example.org/index.html` for the first time.

(a) List in order the exchanges needed until the page is received with HTTP/1.1: protocol, transport, port, role of each exchange. (2 pts)

1. DNS: A record of www.example.org asked to the local resolver, UDP 53. 2. TCP: 3-way handshake (SYN; SYN,ACK; ACK) to port 443. 3. TLS: handshake after the TCP handshake; the server sends its certificate and a common encryption key is agreed. 4. HTTP: GET /index.html, reply 200 OK with the document, encrypted by TLS. With HTTP/1.1 the same TCP connection is then reused for the other objects of the page.

(b) The server presents an X.509 certificate. Under which two conditions does the browser accept it? What does a self-signed certificate allow, and what does it not allow? (1 pt)

It was issued to the site we are connected to, and a CA that we trust has signed it. Self-signed: usable for encryption, not for validating the identity of the site.

(c) What changes with HTTP/3 over QUIC: transport, handshakes, encryption, mobility? (1 pt)

QUIC runs over UDP, implemented in the application layer; the session and TLS handshakes are combined (3-way instead of 6-way); encryption is mandatory; the transport session ID is independent of the ports, which supports mobility; multiplexing is aware of web elements.

(d) Outside the Web, session 1: the course distinguishes IntServ, DiffServ and best effort. How do they differ? Which one describes how the Internet runs, and why? (1 pt)

IntServ reserves resources for individual flows (fine-grained); DiffServ reserves for classes of flows; best effort makes no reservation. The Internet runs best effort: reservation needs an entity that owns the network, grants resources and enforces limits, and the Internet has no single owner.

Marking. (a) 2, 0.5 per exchange in the right order. (b) 1. (c) 1. (d) 0.5 for the three definitions, 0.5 for the Internet and the reason. **Pitfalls.** Putting TLS before TCP; calling QUIC a kernel transport protocol (it is an application layer protocol over UDP); swapping IntServ (per flow) and DiffServ (per class).

1. Session 1: Protocol stack and statistical multiplexing (5 points) inspired by: hta1p p. 3 to 6 and 20; hta5 p. 7

A browser sends an HTTP request; it goes down through TCP, IPv4 and Ethernet.

(a) Name the 7 OSI layers and the 4 TCP/IP layers and match them. Give the PDU name at the transport, network, datalink and physical layers. (1.5 pts)

> OSI Application, Presentation, Session → TCP/IP Application; Transport → Transport; Network → Network; Datalink, Physical → Link. PDUs: Segment (transport), Packet (network), Frame (datalink), Bit (physical); Data at the application layer.

(b) In which order do the headers appear in the transmitted frame? What is encapsulation, and why is the UDP checksum, which covers fields of the IP header, called a layering violation? (1 pt)

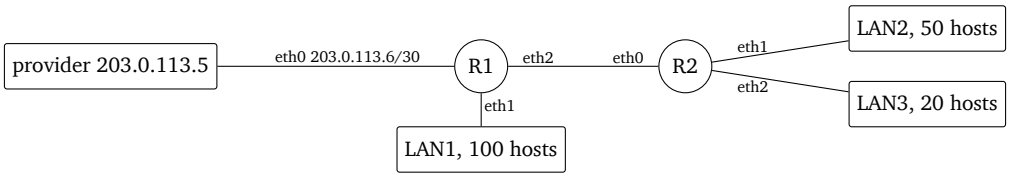
> Ethernet header, IP header, TCP header, HTTP data (Ethernet may also add a footer). Encapsulation: each layer adds its own header (possibly a footer) and only touches its own header. The UDP checksum includes part of the IP header (addresses, length, protocol): one layer depends on another layer's header, which is a layering violation.

(c) Statistical multiplexing: $N = 400$ subscribers share a multiplexer; each is active with probability $p = 2\%$ at a given time, independently. What is the average number of active subscribers? Which distribution does the number X of active subscribers follow, and which approximation does the course use? With Chebyshev's inequality for $k = 3$ (take $\sigma = \sqrt{\lambda}$ for the Poisson distribution), which capacity L (an integer number of links) guarantees a blocking probability $\Pr(X > L)$ of at most 1/9? The exact Poisson computation gives $\Pr(X > L) \approx 0.16\%$ for that L: what do you conclude about Chebyshev? (2.5 pts)

> Average $Np = 8$. X is binomial ($N = 400$, $p = 0.02$); with large N and small p the course uses the Poisson distribution with $\lambda = Np = 8$. $\sigma = \sqrt{8} \approx 2.83$; $L = 8 + 3 \times 2.83 = 16.49$, so $L = \mathbf{17}$ links. Then $\Pr(X > 17) \le \Pr(|X - 8| \ge 3\sigma) \le 1/9 \approx 11\%$. The exact value, about 0.16%, is far smaller: Chebyshev does not depend on the distribution, so it overestimates and over-dimensions the multiplexer.

Marking. (a) 1.5: layers 1, PDUs 0.5. (b) 1: order 0.5, layering violation 0.5. (c) 2.5: mean and distributions 1, $L = 17$ 1, conclusion 0.5. **Pitfalls.** Dimensioning for 400 links; the slide writes $\sigma = \lambda$ (that is the variance), which would give $L = 8 + 24 = 32$: here the statement fixes $\sigma = \sqrt{\lambda}$; rounding 16.49 down to 16.

2. Session 2: Interface addresses and routing tables (5 points) inspired by: 2018 midterm, Q7; 2019 practice set, ex. 1



The company owns the block 192.168.40.0/24 for LAN1, LAN2, LAN3 and the R1-R2 link.

(a) Split the block from its start, largest need first: network/prefix of LAN1, LAN2, LAN3 and of the link. (1.5 pts)

> LAN1 100 hosts: /25 (126); LAN2 50: /26 (62); LAN3 20: /27 (30); link: /30. LAN1 **192.168.40.0/25**, LAN2 **192.168.40.128/26**, LAN3 **192.168.40.192/27**, link **192.168.40.224/30**.

(b) Give an address to every interface of R1 and R2 on these networks, and the default gateway of a PC on LAN3. (1 pt)

> One possible choice: R1 eth1 192.168.40.1/25, R1 eth2 192.168.40.225/30; R2 eth0 192.168.40.226/30, R2 eth1 192.168.40.129/26, R2 eth2 192.168.40.193/27. Default gateway of a LAN3 PC: 192.168.40.193, the interface of R2 on its LAN.

(c) Write the routing tables of R1 and R2 (destination/prefix, next hop or interface), including the default route. (1.5 pts)

R1		R2	
192.168.40.0/25	eth1 (direct)	192.168.40.128/26	eth1 (direct)
192.168.40.224/30	eth2 (direct)	192.168.40.192/27	eth2 (direct)
203.0.113.4/30	eth0 (direct)	192.168.40.224/30	eth0 (direct)
192.168.40.128/26	via 192.168.40.226	0.0.0.0/0	via 192.168.40.225
192.168.40.192/27	via 192.168.40.226		
0.0.0.0/0	via 203.0.113.5		

> R2's default route covers LAN1 and the Internet; an extra line 192.168.40.0/25 via 192.168.40.225 is also correct.

(d) Can R1 replace its routes to LAN2 and LAN3 with the single line 192.168.40.128/25? Follow a packet destined to 192.168.40.240. Why must R1 do NAT on eth0? (1 pt)

> It would not be an exact merge: /26 and /27 differ in size. 192.168.40.128/25 covers 128 to 255: LAN2, LAN3, the link (the directly connected /30 is more specific and wins) but also 228 to 255, which are unused. A packet to 192.168.40.240 would go from R1 to R2; R2 only has its default route and sends it back to R1: a loop until TTL = 0. Better keep the two lines. NAT: 192.168.0.0/16 is a private range, not routable on the Internet; R1 replaces the source address with its public address 203.0.113.6.

Marking. (a) 1.5. (b) 1. (c) 1.5: remote networks via the neighbour's address on the shared link, default route. (d) 0.5 loop, 0.5 NAT. **Pitfalls.** Using as next hop an R2 address that is not on the shared link (192.168.40.129); forgetting the return route on R2; giving LAN2 a /26 that overlaps LAN1.

3. Session 3: DHCP, DNS and ICMP (5 points) inspired by: no past paper; hta3 p. 4 to 17; hta3p p. 18 to 20

A laptop with no network configuration joins a local network; the user then runs `traceroute www.example.org`.

(a) Describe the DHCP exchange: the four messages in order, which ones are broadcast and why, the transport and the ports, three pieces of information supplied besides the address. What happens if the lease is not renewed? (2 pts)

DORA. Discover (client; link layer and IP broadcast: it has no address and does not know the server); Offer (a server offers an address; several servers may answer); Request (broadcast: it notifies all DHCP servers of the chosen offer); Acknowledgement (confirmation). UDP, server port 67, client port 68. TLV options: netmask, default gateway, DNS server (also time server, hostname). The lease is soft state: if it is not renewed it expires; using an expired address is a problem if the server assigns it to another host.

(b) How does the laptop obtain the IPv4 address of www.example.org: which server does it ask and how does it know that server, which record does it request, where can the answer come from? (1.5 pts)

It asks its local DNS resolver, whose address it received from DHCP. It requests the A record (AAAA for IPv6), over UDP port 53. The request is forwarded along the tree, first upward then downward, until a resolver that is authoritative for the name or that has a cached response.

(c) Explain how traceroute discovers the routers on the path: IP header field used, ICMP messages involved. (1.5 pts)

It sends probes with TTL = 1, then 2, 3… Each router decrements the TTL; the one where it reaches 0 drops the packet and sends back ICMP Time Exceeded (type 11), revealing its address. Probes are UDP to a random port by default, ICMP Echo Request with `-I`, TCP SYN with `-T` (tools slides). An Echo Request that reaches the destination gets an Echo Reply (type 0).

Marking. (a) 2: order of messages 1, broadcast and ports 0.5, options and lease 0.5. (b) 1.5. (c) 1.5: TTL 0.75, Time Exceeded 0.75. **Pitfalls.** Thinking Request is unicast; mixing up the local resolver and the authoritative server; saying that the routers answer the probes with Echo Reply.

4. Session 4: One Bellman-Ford iteration (distance vector) (5 points) inspired by: 2017 topic list, no. 15; hta4p p. 3 and 4

Vector from	A	B	C	D	E	F
A (cost 2)	0	4	6	5	8	10
B (cost 7)	4	0	2	1	3	9
C (cost 3)	6	2	0	7	2	4

Router X has three neighbours, A, B and C, with the link costs shown. It has just received their distance vectors (left, destinations other than X).

(a) Compute X's new table from these vectors only: for each destination A to F, distance and next hop, with the formula used. (2.5 pts)

$d_X(Y) = \min_{v \in \{A,B,C\}} \big(c(X,v) + d_v(Y)\big)$, with $d_v(v) = 0$.

Dest.	via A ($2 + d_A$)	via B ($7 + d_B$)	via C ($3 + d_C$)	Distance	Next hop
A	2	11	9	2	A
B	6	7	5	5	C
C	8	9	3	3	C
D	7	8	10	7	A
E	10	10	5	5	C
F	12	16	7	7	C

B is reached via C (5), not over the direct link (7).

(b) Which vector does X advertise next? With split horizon, what does X's advertisement to C contain? (1 pt)

X advertises (A 2, B 5, C 3, D 7, E 5, F 7), without next hops. Split horizon: X does not advertise to C the routes whose next hop is C (B, C, E, F); to C only A 2 and D 7.

(c) Compare distance vector and link state (information sent, recipients, algorithm, convergence), with one protocol of each family. Which protocol is used between autonomous systems, and of which type is it? (1.5 pts)

DV: tell my neighbours who I know (destination, distance), to neighbours only, distributed Bellman-Ford, slow convergence and count to infinity; RIP, IGRP. LS: tell everybody who my neighbours are (LSA), to all known nodes, Dijkstra on the whole topology, faster reconfiguration after a link failure; OSPF, IS-IS. Between ASes: BGP, path vector (whole paths are exchanged), policy routing.

Marking. (a) 2.5: formula 0.5, table 2. (b) 1. (c) 1.5. **Pitfalls.** Keeping the direct link to B by reflex; adding the link cost twice; putting the next hop in the advertisement.

5. Session 5: TCP connection, fast retransmit and window size (5 points) inspired by: 2020 exams; hta5 p. 9, 10; hta5p p. 13, 16

(a) A client opens a TCP connection with initial sequence number $S_c = 7300$; the server picks $S_s = 1200$. Give the flags, Sequence and Acknowledgement numbers of the three handshake segments. Which attack targets the server while it waits for the last segment, and which defense does the course give? (1.5 pts)

SYN (seq 7300); SYN,ACK (seq 1200, ack 7301); ACK (seq 7301, ack 1201). SYN flood: lots of SYN segments with no intention to connect exhaust the connection table, where the server remembers the sequence numbers it has to verify. Defense: SYN cookies; the server does not remember its chosen sequence number but generates it with a cryptographic function and checks the client's ACK with the same function.

(b) As on the slides, numbers count segments. The sender sends segments 40 to 45; segment 42 is lost; the receiver acknowledges every segment it receives. List the ACKs. When does Reno retransmit, which ACK follows the arrival of the retransmitted 42, and what does Reno do to cwnd? (1.5 pts)

40: ACK 41; 41: ACK 42; 43: ACK 42 (1st duplicate); 44: ACK 42 (2nd); 45: ACK 42 (3rd duplicate): fast retransmit of 42. When 42 arrives: **ACK 46**, cumulative, since 43 to 45 were already received. Fast recovery: cwnd/2 and stay in congestion avoidance, no slow start.

(c) A 200 Mbit/s link has a 60 ms RTT. Which window, in octets, is needed to use it fully? Which maximum throughput does the Window Size field alone allow (16 bits, counted in octets)? Which minimum window scale shift is needed? (2 pts)

Window $\geq$ rate $\times$ RTT $= 200 \cdot 10^6 \times 0.060 = 12 \cdot 10^6$ bits $=$ **1.5 MB** (1,500,000 octets). Window Size alone: 65,535 octets per RTT, so $65{,}535 \times 8/0.060 \approx$ **8.74 Mbit/s**, 4.4% of the link. Window scale: $65{,}535 \times 2^4 = 1{,}048{,}560$ is not enough, $65{,}535 \times 2^5 = 2{,}097{,}120 \geq 1{,}500{,}000$: shift **5** (maximum 14).

Marking. (a) 1.5: segments 1, attack and defense 0.5. (b) 1.5: ACK list 0.5, moment of retransmission 0.5, ACK 46 and cwnd 0.5. (c) 2: bandwidth-delay product 1, throughput without the option 0.5, shift 0.5. **Pitfalls.** ACK S_c instead of $S_c + 1$; retransmitting at the first duplicate ACK (that is New Reno); answering ACK 43 after the retransmission; mixing up bits and octets; using the one-way delay instead of the RTT.

6. Session 6: E-mail (5 points)

inspired by: no past paper; hta6p p. 4 to 18

alice@example.org sends, from her mail client, a message with a PDF attachment to bob@example.net, who reads his mail on his phone and on his computer.

(a) Describe the path of the message: agents crossed (MUA, MTA, MDA), protocol and port at each step, and how the MTA of example.org finds the one of example.net. (2 pts)

Alice's MUA submits the message with SMTP on the submission port 465 (formerly 587), with TLS and authentication, to the MTA of example.org. That MTA looks up the MX record of example.net in DNS and relays with SMTP, port 25, to the MTA of example.net. The MDA of example.net (a relay that can store messages) puts it in Bob's mailbox. Bob's MUA retrieves it with IMAP (143, 993 with TLS) or POP3 (110, 995).

(b) Why should Bob use IMAP rather than POP3? (1 pt)

IMAP queries the message database of the server: multiple mailboxes in directories, headers and attachments downloaded separately, browsing without retrieving; both devices see the same state. POP3 is simple retrieval of the inbox only, optionally deleting from the server, no subfolders.

(c) The message format only allows 7-bit ASCII: how is the attachment carried? (1 pt)

MIME converts it to 7-bit ASCII (base64 or quoted-printable encoding), labels it with `Content-Type: application/pdf`, and puts text and attachment in a multipart body.

(d) How can the MTA of example.net check that the message comes from a server authorized for example.org? Name two mechanisms and their principle. (1 pt)

SPF: the MTA proves that it is authorized to send mail for the domain in the From header. DKIM: the sender MTA signs the mail, with its public keys in DNS. (DMARC publishes in DNS how the domain's MTA is set up for DKIM and SPF.)

Marking. (a) 2: chain of agents 1, ports 0.5, MX 0.5. (b) 1. (c) 1. (d) 0.5 per mechanism. **Pitfalls.** Retrieving mail with SMTP; saying that Alice's MUA contacts Bob's server directly; mixing up DKIM (signature) and SPF (authorization of the sending MTA).